Abdiel Rodríguez Lara

Software Engineer

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Summary

AI Engineer with a strong background in full-stack software development and production-grade AI systems. Specialized in designing and deploying Retrieval-Augmented Generation (RAG) architectures, multi-agent systems, and AI-driven automation workflows. Experienced in building scalable backend integrations in Python, orchestrating LLM pipelines, and implementing federated learning and cybersecurity techniques for distributed environments.
Previously developed robust enterprise web applications using Spring Boot, Nest.js, Angular, and Vue.js, and brings solid QA expertise in web accessibility and functional testing. Combines AI architecture, backend engineering, and system-level thinking to deliver reliable, production-ready solutions.
Currently based in Cuba, relocating permanently to Canada in August 2026.

Experience

02/2026 – Present
Marianao, La Habana, Cuba

AIKoders
AI Engineer & Full Stack Developer

- Developing a "Guest Concierge" AI system using n8n and RAG architectures to automate rental property management. The agent autonomously queries Supabase databases to resolve guest inquiries and provide local recommendations, while intelligently escalating critical needs directly to property owners.
- Orchestrating an autonomous multi-agent marketing system using OpenClaw. Designed specialized sub-agents and skills to conduct competitor analysis, extract industry keywords, and generate comprehensive monthly social media campaigns (IG, FB) utilizing generative AI models like Ideogram for aesthetic imagery and Kling for realistic videos.
- Engineering a high-performance Cuban address autocomplete system from scratch utilizing a modern stack with FastAPI and Next.js.
- Designing a complex geospatial database architecture with PostgreSQL, PostGIS, and Meilisearch to efficiently parse, search, and resolve the highly specific formatting nuances of Cuban addresses (including cross-streets, neighborhoods, and municipalities).
- Technologies: FastAPI, Next.js, PostgreSQL, PostGIS, Meilisearch, Supabase, n8n, OpenClaw, RAG, Ideogram, Kling, Multi-Agent Systems, Python, TypeScript, DigitalOcean.

09/2025 – Present

Centro de Ingeniería Genética y Biotecnología (CIGB)
Freelance Bioinformatics Software Engineer

- Designed and implemented a quantitative scoring algorithm to evaluate the reliability of protein–protein interactions between Dengue virus components and the human LRP1 protein. The model applied weighted criteria based on experimental evidence type, interaction frequency, and dataset characteristics to produce normalized confidence scores.

- Performed structured analysis of biological interaction datasets, modeling heterogeneous experimental data into a reproducible scoring framework suitable for scientific interpretation.
- Developed a web-based analytical platform enabling advanced querying by protein name, gene symbol, UniProt ID, and cell type, integrating the custom scoring engine into the application's search core.
- Optimized data filtering and retrieval processes to support efficient exploration of complex biological interaction networks.

Technologies: Angular 21, TypeScript, Bioinformatics Data Processing, Algorithm Design, REST APIs, Python

Github Repositories: https://github.com/Erick0330/Score-algorithm, https://github.com/Erick0330/data_base_lrp1_web_app

Live Site: https://data-base-lrp1-web-app.onrender.com/

01/2026 – 02/2026
La Habana, Cuba

Guajiritos

AI Engineer | RAG & Automation Specialist

- Designing and implementing Retrieval-Augmented Generation (RAG) architectures to enhance LLM outputs using proprietary and structured business data.
- Building scalable automation workflows in n8n, integrating APIs, vector databases, and AI models to optimize internal business processes.
- Developing production-ready backend and integration layers in Python to deploy AI-powered features into operational systems. Implementing multi-agent AI systems and MCP-based orchestration to coordinate complex task execution and dynamic tool usage.

Technologies: Python, n8n, LangChain, Vector Databases, REST APIs, RAG Architectures, Multi-Agent Systems, MCP

11/2025 – 01/2026
La Habana, Cuba

INTRUST SECURITY

Full Stack Developer

- Developed a comprehensive vulnerability management platform for a cybersecurity company, building both backend and frontend layers using Spring Boot and Angular.
- Designed and optimized relational database schemas with PostgreSQL and managed versioned database migrations using Liquibase.
- Built automated tools to scan and detect vulnerabilities across specific domains, improving the efficiency of security assessments. Integrated multiple security scanning tools into a unified dashboard for centralized monitoring and real-time risk analysis.

Technologies: Spring Boot, Angular, PostgreSQL, Liquibase, REST APIs, Go

01/2025 – 11/2025
La Habana, Cuba

A11ySolutions

Web Accessibility Auditor/Tester

- Worked in the Quality Assurance sector, initially performing functional testing for web applications, including high-visibility platforms such as BillShark.com, a company associated with Mark Cuban. Responsibilities included validating user flows, detecting functional defects, and ensuring consistent performance across supported environments.
- Later transitioned to the role of Web Accessibility Auditor, leading accessibility evaluation projects for Crownpeak. Identified, documented, and reported WCAG compliance issues across multiple enterprise-level websites, producing clear, testable issue reports and collaborating with development teams to guide remediation.

Technologies and skills: Functional testing, WCAG 2.2, screen readers (NVDA/JAWS/VoiceOver), accessibility auditing, issue documentation, regression testing, cross-browser testing.

10/2024 – 05/2025
La Habana, Cuba

ALASolutions
Full-Stack web developer

- Worked on two of the company's most innovative projects as a Full Stack Software Engineer. Contributed to a real-time GPS vehicle monitoring system based on Traccar(https://github.com/traccar/traccar.git), fixing bugs, improving backend performance, and enhancing overall usability. Also helped build a simplified version for Gaviota by redesigning the UI for non-technical users.
- Additionally, participated in the initial design and architecture of an intelligent charging system developed from scratch. Collaborated on core components, user flows, and API integrations, including ETECSA's Transfermóvil for secure payments across distributed charging stations in Havana.

Technologies used: Nest.js, Spring Boot, Angular, Vue.js, Java, TypeScript, REST APIs, UX/UI optimization, API integrations.
GitHub Repositories: https://github.com/Erick0330/Cargando.git
https://github.com/Nelsonma98/SmartCharger-back.git

08/2024 – 11/2024
Playa, La Habana

ETECSA
Backend Developer

- Contributed to the development of a digital transformation consulting application for businesses. Responsible for designing and managing the database as well as developing the backend system. Collaborated with a team of three developers to deliver a robust solution. The system generates a variety of graphs and tables to assess the level of digital transformation within a company, based on responses to a structured set of questions organized by dimensions, perspectives, and areas.

Technologies used: Java, SQLite.
GitHub Repository: https://github.com/Erick0330/TETRDIG.git

Education

Marianao, La Habana, Cuba

Software Engineering (Currently studying)
Ciudad Universitaria José Antonio Echeverría (CUJAE)
Marianao, La Habana, Cuba *4th year student — Expected graduation: 2026*

Skills

Programming Languages ●●●●○
Python, Java, JavaScript, TypeScript , C, C++

AI Orchestration & Automation ●●●●○
n8n Workflow Automation, Multi-Agent Systems, MCP (Model Context Protocol), Tool Integration & API Orchestration, AI-driven business process automation, OpenClaw

Vector Databases ●●●●○
Vector search, Similarity search, Embedding storage, Qdrant, Supabase

Backend Development ●●●●○
API design, microservices, authentication workflows

Version Control ●●●○○
Git, branching workflows, code reviews

Accessibility Standards ●●●●○
WCAG 2.2, ARIA techniques, semantic HTML

QA Testing ●●●○○
Functional testing, regression testing, cross-browser testing

Neural Networks ●●●○○
Model design, training, evaluation, optimization

General ML Skills ●●●○○
Data preprocessing, experiment tracking, reproducibility

Problem-Solving ●●●●●
Strong debugging and analytical skills

Frameworks & Libraries ●●●●○
PyTorch, TensorFlow, Flax, FLEXible, LangChain, Spring Boot, Angular, Vue.js, Nest.js, REST APIs, FastAPI, Next.js

LLM & RAG Systems ●●●●○
Retrieval-Augmented Generation (RAG), Prompt Engineering, Context Window Optimization, Embedding Models, Chunking Strategies, Evaluation of LLM outputs, Generative AI (Image & Video)

Frontend Development ●●●●○
HTML, CSS, responsive design, reusable components

Databases ●●●●○
PostgreSQL, MySQL, SQLite, PostGIS, Meilisearch, Redis

Performance & Debugging ●●●○○
Performance optimization, debugging, software design standards

Accessibility Audit Skills ●●●●○
Manual audits, keyboard navigation checks, screen reader validation, Wave, Axe

Issue Documentation
Clear defect writing, reproducible steps, remediation guidance

Federated Learning ●●●○○
Client-server orchestration, secure aggregation, robustness techniques, attack methods

Teamwork & Collaboration ●●●●●
Effective communication, cross-functional teamwork

Documentation ●●●●○
API documentation, technical writing, structured reporting

Languages

Spanish
First Language

English
Advanced(C1)

Academic projects

01/2026 – 01/2026

Real-Time Drowsiness Detection System
Computer Vision Engineer

- Developed a real-time driver drowsiness detection system using YOLO (Ultralytics) trained on a custom Roboflow dataset and fine-tuned in Google Colab. The model detects eye states (open/closed) and yawning to infer fatigue levels.

- Built a desktop monitoring application in Python integrating OpenCV for live video capture and inference, applying frame-by-frame object detection to analyze driver behavior in real time.
- Implemented a state-based logic system to prevent false positives, including configurable sensitivity thresholds, debounce mechanisms for yawning detection, and temporal validation for sleep events.
- Designed an interactive GUI using CustomTkinter featuring live video feed, visual bounding boxes, fatigue progress indicators, event logging, and real-time analytics dashboards with Matplotlib.
- Integrated an auditory alert system that triggers warnings when drowsiness thresholds are exceeded, simulating a production-ready safety monitoring tool.

Technologies: Python, YOLO (Ultralytics), OpenCV, Roboflow, Google Colab, CustomTkinter, Matplotlib, Computer Vision
GitHub Repository: https://github.com/Erick0330/Real-Time-Drowsiness-Detection-System

Comparative analysis between federated and traditional learning
AI Research Developer
Conducted an academic research project to evaluate the effectiveness of Federated Learning (FL) in tasks traditionally solved by centralized machine learning models. Implemented federated training workflows to compare performance, privacy guarantees, and robustness across ten diverse datasets from the UCI Machine Learning Repository.
Technologies: Python, Federated Learning, Flex, Machine Learning, UCI Datasets
GitHub Repository: https://github.com/AdrianRodriguezJorge/test_FL.git

Tic-Tac-Toe Game
Full Stack Developer
Designed and developed a web-based Tic-Tac-Toe game for two players, featuring a multiversal travel theme that alters the color palette, icons, images, and background music upon a win. Integrated statistics to track player wins and introduced a secret mode unlocked after 5 ties.
Technologies: Vue.js
Live Site: https://tic-tac-toe-anime.onrender.com
GitHub Repository: https://github.com/Erick0330/tic-tac-toe.git

Mass Spectrometry Workshop Web Platform
Frontend Developer & UI Designer
Designed and developed a fully responsive web platform to promote a Mass Spectrometry workshop, showcasing strong frontend engineering and UI/UX design skills. Built the entire application using Vue.js, including custom layouts, reusable components, and a clean visual aesthetic tailored for scientific content presentation. Implemented a modern interface optimized for performance and cross-device compatibility, ensuring an intuitive user experience for participants and visitors.
Technologies: Vue.js, HTML, CSS, responsive design, UI/UX
GitHub Repository: https://github.com/Erick0330/curso_espectrometria_masa

Animal Shelter Management Application
Software Architect
Developed an application to automate the management of an animal shelter, including CRUD functionality for animals, staff, and their subcategories. Integrated a PDF reporting system using JasperReports and implemented role-based authentication and authorization for enhanced security. Responsible for designing the database schema, implementing backend functionalities, and managing the backend structure effectively.
Technologies: Java, JasperReports, PostgreSQL.
GitHub Repository:

https://github.com/Erick0330/RefugioVSC.git

iMessage Simulator
Full Stack Developer
Developed an offline email system simulation, allowing users to send messages to different recipients and test the functionality of email reception. Implemented a Circular Queue data structure from scratch to efficiently manage emails and included a graphical interface for a user-friendly experience. Features include a recycle bin and a text file system for data persistence across sessions. Detailed instructions on server connection and message-sending mechanisms are provided in the GitHub repository.
Technologies: C++
GitHub: https://github.com/Erick0330/iMessageSimulator.git